AI DEVELOPMENT INTERNSHIP PROGRAM

AT JARGON HANDLERS SOFTWARE SOLUTIONS PVT LTD

AN INTERNSHIP PROJECT REPORT

Submitted by

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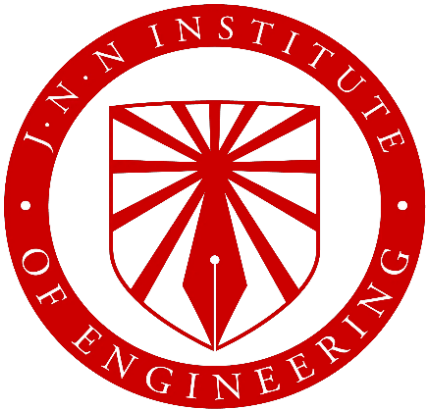
in partial fulfilment for the award of the degree of

of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE & DATA SCIENCE



J.N.N INSTITUTE OF ENGINEERING

(An Autonomous Institute, Affiliated to Anna University, Chennai)

AUGUST – 2026

VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE:

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instils in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.

VISION OF THE DEPARTMENT:

To impart quality-education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT:

- **DM 1:** To enhance the knowledge of the students with most recent advancements and refresh their insights in the field of Artificial Intelligence and Data Science.
- **DM 2:** To equip the students with strong fundamental concepts, analytical capability, programming and problem-solving skills.
- **DM 3:** To make the students industry ready and to enhance their employability through training, internships and real-time projects.
- **DM 4:** To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry.

PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Fore sight, Insight, hind sight from data for solving business and engineering problems
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.



BONAFIDE CERTIFICATE

Certified that this internship project report " AI DEVELOPMENT INTERNSHIP PROGRAM AT JARGON HANDLERS SOFTWARE SOLUTIONS PVT LTD " is the bonafide work of "M. SUSHIT" (Reg No: 110725102253) who carried out the project work under my supervision.

Signature of the HOD with date

Dr. P. Nagarajan
Professor and Head,
Department of Artificial Intelligence
& Data Science Engineering,
J.N.N Institute of Engineering,
Chennai

Signature of the Supervisor with date

Ms. M. Gayathridevi
Project Supervisor,
Department of Artificial Intelligence
& Data Science Engineering,
J.N.N Institute of Engineering,
Chennai

Certified that the above students have attend of viva voce during the exam held on…….

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

First and foremost, we express our sincere gratitude to our Beloved FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L., and also heartfelt thanks to our VICE CHAIRMAN Mr. NAVEEN JAYAHANDRAN M.S., for providing necessary facilities for the completion of our project.

We thank our PRINCIPAL Dr. G. MOHANBABU, Ph.D., for the encouragement provided with the congenial environment essential for us to complete the project.

We take this opportunity to express our gratitude to our HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving us continuous encouragement for completing our project.

We extend our sincere thanks to Jargon Handlers Software Solutions Pvt Ltd and its Director - Operations, Mr. Madasamy Shanmugavel, for giving us this opportunity and for the guidance and support provided throughout this internship.

Finally, we would like to thank our parents and friends for their constant support and encouragement.

ABSTRACT

Every web application, from a simple personal page to a modern AI chatbot, is built on HTML and CSS. HTML gives a page its structure and content, while CSS controls how that content looks and how it adapts to different screens. This report documents the work carried out during the AI Development Internship Program at Jargon Handlers Software Solutions Pvt Ltd, Chennai, completed between 1st August 2026 and 31st August 2026, as an AI Development Intern.

As part of the internship, five connected front-end tasks were designed, coded and tested: a personal portfolio page, a styled registration form, a responsive card layout built with CSS Grid, a landing page with a sticky navigation bar and hover effects, and an AI chatbot interface. Together these tasks covered semantic HTML5, the CSS box model, Flexbox, CSS Grid, media queries, transitions and form styling. The code for each task is presented in the report along with its rendered output.

Overall, the internship strengthened technical, problem-solving and design skills and gave a clear understanding of how the user interface layer of an AI application is built, which is the part of the system that people actually see and use.

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CHAPTER 1
INTRODUCTION

1.1 Overview

The World Wide Web is the most widely used platform for delivering software. Every web application, from a simple personal page to a modern AI chatbot, is built on two core technologies: HyperText Markup Language (HTML), which defines the structure and content of a page, and Cascading Style Sheets (CSS), which controls its appearance and layout. Even intelligent systems that rely on complex models need a clear, usable and responsive interface so that people can interact with them.

This report presents five connected front-end tasks completed during the AI Development Internship Program at Jargon Handlers Software Solutions Pvt Ltd, from 1st August 2026 to 31st August 2026, as an AI Development Intern. The tasks progress from a simple portfolio page (Task 1) and a styled form (Task 2) to a responsive card layout (Task 3), a landing page with interactive effects (Task 4), and finally an AI chatbot interface (Task 5).

1.2 Internship Details

Item	Details
Organisation	Jargon Handlers Software Solutions Pvt Ltd
Programme	AI Development Internship Program
Role	AI Development Intern
Duration	1st August 2026 to 31st August 2026
Certificate issued by	Mr. Madasamy Shanmugavel, Director - Operations (dated 08/09/2026)
Student	M. Sushit (Reg No: 110725102253), B.Tech Artificial Intelligence & Data Science

1.3 Objectives

- Build a personal portfolio page using semantic HTML5 elements and CSS Flexbox.
- Design and style a registration form with proper labels, input types and focus effects.
- Create a responsive card layout using CSS Grid and media queries that adapts to desktop, tablet and mobile screens.
- Develop a landing page with a sticky navigation bar, gradient hero section and hover transitions.
- Design an AI chatbot interface with message bubbles and an input bar using Flexbox.

1.4 Overview of the Five Tasks

Task	Title	Key Concepts
Task 1	Personal Portfolio Page	Semantic HTML5, box model, Flexbox, box-shadow
Task 2	Styled Registration Form	Form elements, input types, :focus, transitions
Task 3	Responsive Card Layout	CSS Grid, media queries, hover transform
Task 4	Landing Page with Navigation Bar	Sticky navbar, gradients, hover effects
Task 5	AI Chatbot Interface	Flexbox column layout, chat bubbles, input bar

CHAPTER 2
LITERATURE REVIEW

This chapter gives the background of the technologies and tools used in the five tasks.

2.1 HyperText Markup Language (HTML)

HTML is the standard markup language for creating web pages. A page is described using elements written as tags, such as headings, paragraphs, lists, images and links. HTML5 introduced semantic elements, including header, nav, main, section and footer, which describe the meaning of each part of a page. This makes pages easier to read, to maintain and to navigate for assistive technologies and search engines.

HTML also provides form elements such as form, input, label, select, textarea and button for collecting information from users. Input types such as text, email and tel, and attributes such as required and placeholder, give basic validation and guidance without any extra code.

2.2 Cascading Style Sheets (CSS)

CSS describes how HTML elements are presented. A CSS rule consists of a selector and a set of declarations. Selectors can target elements, classes, identifiers and states such as :hover and :focus. The cascade, specificity and inheritance rules decide which declaration is applied when several rules match the same element.

CSS controls colours, typography, backgrounds, gradients, borders, shadows and spacing. The transition property animates changes smoothly, for example when a button changes colour on hover. In these tasks, the CSS is written inside a style element in the head of each page so that every program is a single, self-contained file.

2.3 CSS Layout: Box Model, Flexbox and Grid

Every element is treated as a rectangular box made up of content, padding, border and margin. Setting box-sizing to border-box makes the width of an element include its padding and border, which keeps layouts predictable.

Flexbox is a one-dimensional layout model that arranges items in a row or a column and controls alignment, spacing and wrapping. CSS Grid is a two-dimensional layout model in which rows and columns are defined together; the repeat() function, the fr unit and the gap property make it simple to build card layouts. The position: sticky value keeps an element, such as a navigation bar, fixed at the top while the page scrolls.

2.4 Responsive Web Design

Responsive web design makes one page work well on screens of different sizes. It relies on the viewport meta tag, flexible layouts, relative units and CSS media queries. A media query applies a set of rules only when a condition, such as a maximum screen width, is true. This allows a four-column grid on a desktop to become two columns on a tablet and one column on a phone.

2.5 Tools and Their Purpose

Tool	Purpose
HTML5	Structure and content of every page
CSS3	Styling, layout, animation and responsiveness
Code editor (e.g. Visual Studio Code)	Writing and organising the source files
Web browser (Chrome / Edge)	Rendering the pages and viewing the output
Browser developer tools	Inspecting elements and testing different screen sizes

CHAPTER 3
METHODOLOGY

3.1 Overall Approach

The internship followed a progression of five tasks, each building on the skills of the one before. Every task was carried out in the same six steps: define the requirement, sketch the page structure, write the HTML, write the CSS, test the page in a browser at different window sizes, and refine the result.

All five programs start from the same HTML5 skeleton shown below. It declares the document type and language, sets the character encoding, adds the viewport meta tag needed for responsive pages, and holds the CSS in a style element.

Listing 3.1 – Basic HTML5 page skeleton used in all tasks

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Page Title</title>
  <style>
    /* CSS rules are written here */
  </style>
</head>
<body>
  <!-- Page content (header, main, footer) goes here -->
</body>
</html>
```

Each program also begins with a small CSS reset. It removes the default browser margins and padding and applies border-box sizing to every element, so that widths behave the same way in all browsers.

Listing 3.2 – CSS reset used at the top of every stylesheet

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: Arial, sans-serif;
}
```

3.2 Task 1 – Portfolio Page Design

The portfolio page is divided into three semantic parts: a header with the name and role, a main area with the About Me, Skills and Education sections, and a footer. Each section is a white card with rounded corners and a soft box-shadow on a light background. The skills are displayed as rounded tags inside a Flexbox container with wrapping enabled, so they flow onto a new line when space runs out.

3.3 Task 2 – Form Design

The registration form is centred on the page using Flexbox on the body. Every field has a label connected to its input through the for and id attributes, which makes the form easier to use and more accessible. Text, email and phone inputs, a drop-down list, radio buttons and a text area cover the main form controls. A :focus rule changes the border colour and adds a glow, and a transition makes the change smooth.

3.4 Task 3 – Responsive Card Layout Design

The service cards are placed in a CSS Grid container with four equal columns. Two media queries reduce the grid to two columns on screens up to 800 px wide and to a single column on screens up to 480 px wide. The core of the layout is shown below.

Listing 3.3 – Grid layout and media queries for the card layout

```
.grid {
  max-width: 1000px;
  margin: auto;
  display: grid;
  grid-template-columns: repeat(4, 1fr);   /* four equal columns */
  gap: 18px;
}
@media (max-width: 800px) {
  .grid { grid-template-columns: repeat(2, 1fr); }   /* tablet */
}
@media (max-width: 480px) {
  .grid { grid-template-columns: 1fr; }              /* mobile */
}
```

3.5 Tasks 4 and 5 – Landing Page and Chatbot Design

The landing page uses a sticky navigation bar built with Flexbox and justify-content: space-between, so that the logo stays on the left and the links on the right. A linear-gradient background gives the hero section depth, and the button and links respond to the mouse with transitions.

The chatbot interface is a fixed-size card arranged as a Flexbox column with three parts: a title bar, a scrollable message area and an input bar. Bot and user messages are rounded bubbles aligned to the left and to the right using align-self, with different background colours so that the two speakers are easy to tell apart.

CHAPTER 4
IMPLEMENTATION

This chapter presents the complete code for each of the five tasks together with the output obtained when the page is opened in a web browser.

4.1 Task 1: Personal Portfolio Page

Concepts used: semantic HTML5 elements, CSS box model, Flexbox, box-shadow, border-radius.

4.1.1 HTML and CSS Code

File: task1_portfolio.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Portfolio - M. Sushit</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }
  body { font-family: Arial, sans-serif; background: #f4f6fb; color: #222;
         line-height: 1.6; }
  header { background: #1f3a93; color: #fff; text-align: center; padding: 40px 20px; }
  header h1 { font-size: 34px; }
  header p { opacity: 0.9; margin-top: 6px; }
  main { max-width: 900px; margin: 30px auto; padding: 0 20px; }
  section { background: #fff; border-radius: 10px; padding: 22px; margin-bottom: 20px;
            box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1); }
  h2 { color: #1f3a93; border-bottom: 2px solid #1f3a93; padding-bottom: 6px;
       margin-bottom: 12px; }
  ul { margin-left: 20px; }
  .skills { display: flex; flex-wrap: wrap; gap: 10px; }
  .skills span { background: #e3e9ff; color: #1f3a93; padding: 6px 14px;
                 border-radius: 20px; font-weight: bold; }
  footer { text-align: center; padding: 18px; background: #222; color: #ddd; }
</style>
</head>
<body>
  <header>
    <h1>M. Sushit</h1>
    <p>B.Tech Artificial Intelligence &amp;amp; Data Science | AI Development Intern</p>
  </header>
  <main>
    <section>
      <h2>About Me</h2>
      <p>I am a final-year B.Tech student who completed an AI Development internship
         at Jargon Handlers Software Solutions Pvt Ltd. I enjoy turning ideas into
         clean, responsive web pages.</p>
    </section>
    <section>
      <h2>Skills</h2>
      <div class="skills">
        <span>HTML5</span><span>CSS3</span><span>Flexbox</span>
        <span>CSS Grid</span><span>Responsive Design</span><span>Python</span>
      </div>
    </section>
    <section>
      <h2>Education</h2>
      <ul>
        <li>B.Tech - Artificial Intelligence &amp;amp; Data Science</li>
        <li>J.N.N Institute of Engineering, Chennai (Anna University)</li>
      </ul>
    </section>
  </main>
  <footer>&copy; 2026 M. Sushit | AI Development Internship Portfolio</footer>
</body>
</html>
```

4.1.2 Explanation

- The header, main and footer elements divide the page into meaningful regions.
- Each section is styled as a card using padding, border-radius and box-shadow.
- The .skills container uses display: flex with flex-wrap and gap to lay out the skill tags.

4.1.3 Output

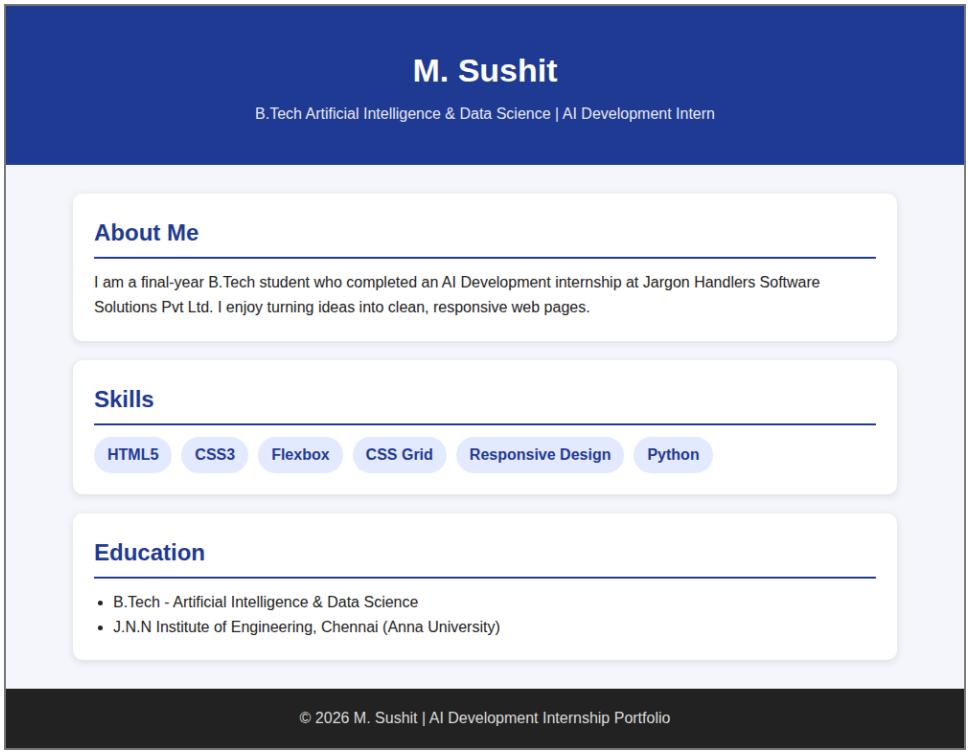


Fig. 4.1 – Output of the personal portfolio page

4.2 Task 2: Styled Registration Form

Concepts used: form controls, label and input association, input types, :focus pseudo-class, transitions.

4.2.1 HTML and CSS Code

File: task2_form.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Workshop Registration Form</title>
<style>
  * { box-sizing: border-box; font-family: Arial, sans-serif; }
  body { background: linear-gradient(135deg, #667eea, #764ba2); min-height: 100vh;
         display: flex; justify-content: center; align-items: center; padding: 20px; }
  form { background: #fff; width: 100%; max-width: 480px; padding: 28px;
         border-radius: 12px; box-shadow: 0 8px 24px rgba(0, 0, 0, 0.25); }
  h1 { text-align: center; color: #4a3f9f; margin-bottom: 18px; font-size: 24px; }
  label { display: block; font-weight: bold; margin: 12px 0 5px; color: #333; }
  input[type="text"], input[type="email"], input[type="tel"], select, textarea {
    width: 100%; padding: 10px; border: 1.5px solid #bbb; border-radius: 6px;
    font-size: 15px; transition: border-color 0.3s, box-shadow 0.3s; }
  input:focus, select:focus, textarea:focus { outline: none; border-color: #764ba2;
    box-shadow: 0 0 6px rgba(118, 75, 162, 0.5); }
  .options label { display: inline-block; font-weight: normal; margin-right: 14px; }
  button { width: 100%; margin-top: 18px; padding: 12px; border: none; border-radius: 6px;
           background: #764ba2; color: #fff; font-size: 16px; cursor: pointer; }
  button:hover { background: #5b3a82; }
```

```
</style>
</head>
<body>
  <form>
    <h1>AI Workshop Registration</h1>
    <label for="name">Full Name</label>
    <input type="text" id="name" placeholder="Enter your name" required>
    <label for="email">Email</label>
    <input type="email" id="email" placeholder="you@example.com" required>
    <label for="phone">Phone Number</label>
    <input type="tel" id="phone" placeholder="10-digit mobile number">
    <label for="track">Preferred Track</label>
    <select id="track">
      <option>Web Development</option>
      <option>Machine Learning</option>
      <option>Data Analysis</option>
    </select>
    <label>Mode of Attendance</label>
    <div class="options">
      <label><input type="radio" name="mode" checked> Online</label>
      <label><input type="radio" name="mode"> Offline</label>
    </div>
    <label for="msg">Message</label>
    <textarea id="msg" rows="3" placeholder="Anything you would like to tell
us"></textarea>
    <button type="submit">Register Now</button>
  </form>
</body>
</html>
```

4.2.2 Explanation

- The body uses Flexbox to centre the form both horizontally and vertically over a gradient background.
- Each label is linked to its input with the for and id attributes, and required marks the compulsory fields.
- The input:focus rule highlights the active field, and transition makes the change smooth.

4.2.3 Output

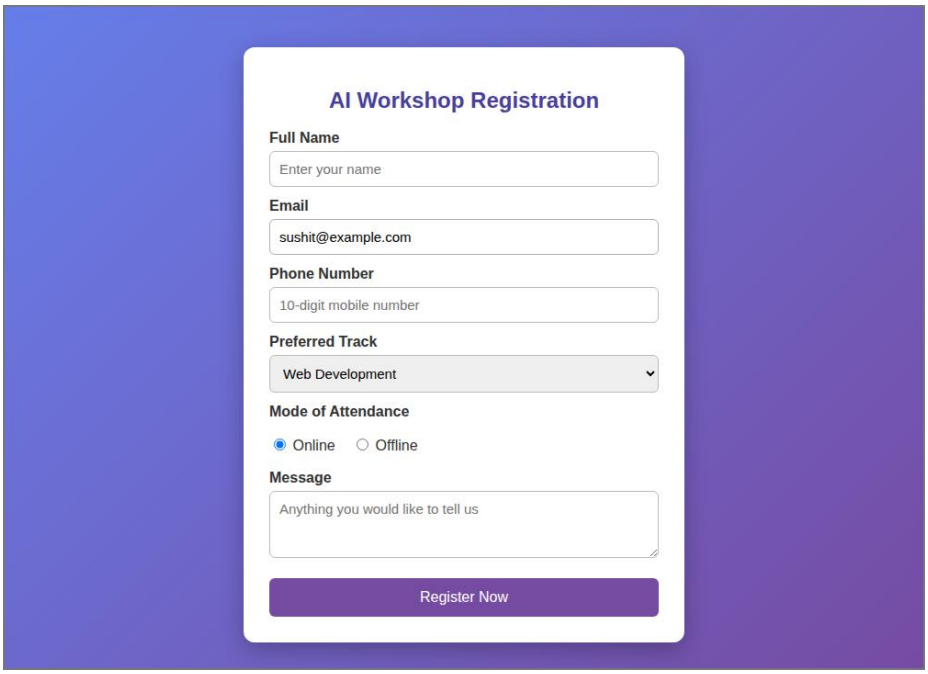


Fig. 4.2 – Registration form with the Email field in focus

4.3 Task 3: Responsive Card Layout Using CSS Grid

Concepts used: CSS Grid, repeat() and fr units, media queries, hover transform and transition.

4.3.1 HTML and CSS Code

File: task3_cards.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>AI Services - Responsive Cards</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; font-family: Arial, sans-serif; }
  body { background: #eef2f7; padding: 30px 20px; }
  h1 { text-align: center; color: #12355b; margin-bottom: 6px; }
  .sub { text-align: center; color: #555; margin-bottom: 26px; }
  .grid { max-width: 1000px; margin: auto; display: grid;
          grid-template-columns: repeat(4, 1fr); gap: 18px; }
  .card { background: #fff; border-radius: 12px; padding: 22px 16px; text-align: center;
          border-top: 5px solid #2e86de; box-shadow: 0 3px 10px rgba(0, 0, 0, 0.1);
          transition: transform 0.3s, box-shadow 0.3s; }
  .card:hover { transform: translateY(-8px); box-shadow: 0 10px 20px rgba(0, 0, 0, 0.18); }
  .card:nth-child(2) { border-top-color: #e67e22; }
  .card:nth-child(3) { border-top-color: #27ae60; }
  .card:nth-child(4) { border-top-color: #8e44ad; }
  .icon { font-size: 34px; margin-bottom: 10px; }
  .card h3 { color: #12355b; margin-bottom: 8px; }
  .card p { color: #555; font-size: 14px; line-height: 1.5; }
  @media (max-width: 800px) { .grid { grid-template-columns: repeat(2, 1fr); } }
  @media (max-width: 480px) { .grid { grid-template-columns: 1fr; } }
</style>
</head>
<body>
  <h1>Our AI Services</h1>
  <p class="sub">Resize the window to see the layout adapt</p>
  <div class="grid">
    <div class="card"><div class="icon">&#128172;</div><h3>Chatbots</h3>
      <p>Conversational assistants that answer customer questions instantly.</p></div>
    <div class="card"><div class="icon">&#128202;</div><h3>Data Analysis</h3>
      <p>Clean, explore and visualise data to support better decisions.</p></div>
    <div class="card"><div class="icon">&#128065;</div><h3>Computer Vision</h3>
      <p>Models that recognise objects and patterns in images.</p></div>
    <div class="card"><div class="icon">&#128214;</div><h3>Language AI</h3>
      <p>Text summarising, translation and sentiment analysis tools.</p></div>
  </div>
</body>
</html>
```

4.3.2 Explanation

- grid-template-columns: repeat(4, 1fr) creates four equal columns, and gap adds space between the cards.
- The :nth-child selectors give each card a different top border colour.
- The two @media rules switch the layout to two columns and then to one column on smaller screens.

4.3.3 Output

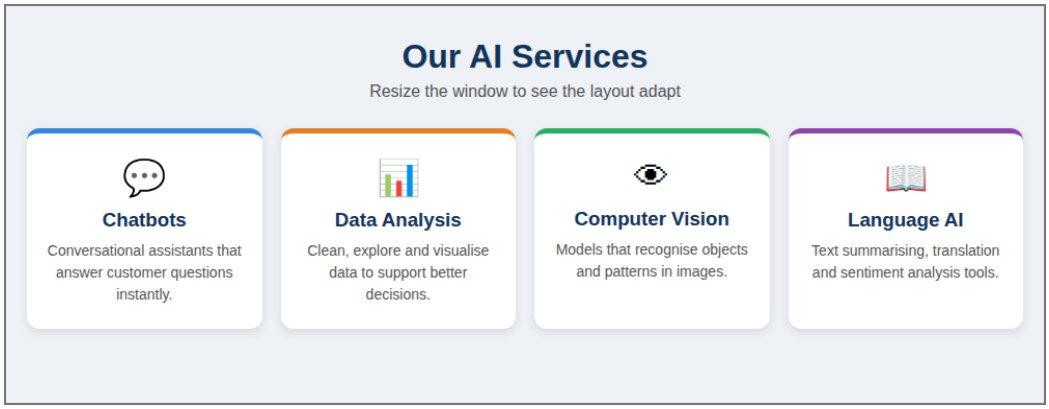


Fig. 4.3 – Card layout on a desktop screen (four columns)

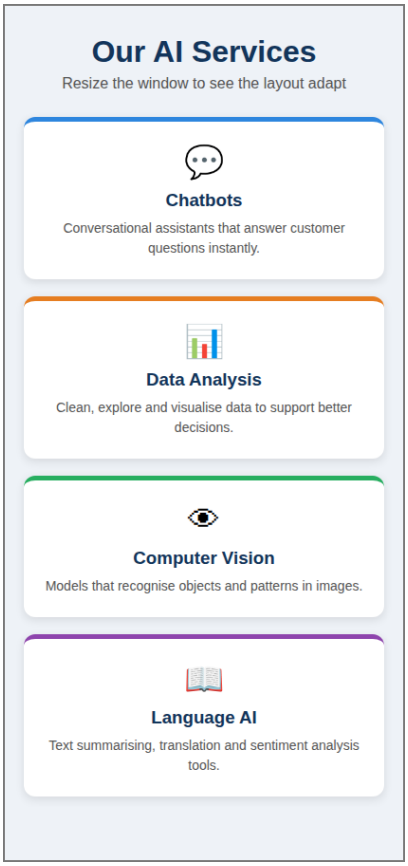


Fig. 4.4 – The same page on a mobile-width screen (one column)

4.4 Task 4: Landing Page with Navigation Bar

Concepts used: sticky positioning, Flexbox, linear-gradient, hover effects, transitions.

4.4.1 HTML and CSS Code

File: task4_landing.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>NeuraLearn - Landing Page</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; font-family: Arial, sans-serif; }
  nav { position: sticky; top: 0; display: flex; justify-content: space-between;
        align-items: center; padding: 16px 40px; background: #0b1d3a; }
```

```
  .logo { color: #4fd1c5; font-size: 22px; font-weight: bold; }
  nav ul { display: flex; list-style: none; gap: 28px; }
  nav a { color: #fff; text-decoration: none; padding-bottom: 4px;
          border-bottom: 2px solid transparent; transition: 0.3s; }
  nav a:hover { color: #4fd1c5; border-bottom-color: #4fd1c5; }
  .hero { text-align: center; color: #fff; padding: 90px 20px;
          background: linear-gradient(135deg, #0b1d3a, #1e5aa8); }
  .hero h1 { font-size: 42px; margin-bottom: 14px; }
  .hero p { font-size: 18px; max-width: 560px; margin: 0 auto 28px; opacity: 0.92; }
  .btn { display: inline-block; background: #4fd1c5; color: #0b1d3a; padding: 13px 32px;
         border-radius: 30px; font-weight: bold; text-decoration: none;
         transition: transform 0.3s, background 0.3s; }
  .btn:hover { background: #fff; transform: scale(1.08); }
  .features { display: flex; justify-content: center; gap: 40px; padding: 36px 20px;
              background: #f3f7fb; text-align: center; }
  .features div { width: 200px; }
  .features h3 { color: #0b1d3a; margin-bottom: 6px; }
  .features p { color: #555; font-size: 14px; }
</style>
</head>
<body>
  <nav>
    <div class="logo">NeuraLearn</div>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">Courses</a></li>
      <li><a href="#">About</a></li>
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>
  <section class="hero">
    <h1>Learn Artificial Intelligence</h1>
    <p>Beginner-friendly lessons that take you from the basics of data
       to building your first intelligent application.</p>
    <a class="btn" href="#">Get Started</a>
  </section>
  <section class="features">
    <div><h3>Hands-on Labs</h3><p>Practise with real exercises after every
lesson.</p></div>
    <div><h3>Expert Mentors</h3><p>Get guidance from experienced industry
mentors.</p></div>
    <div><h3>Certificates</h3><p>Earn a certificate when you finish a course.</p></div>
  </section>
</body>
</html>
```

4.4.2 Explanation

- position: sticky keeps the navigation bar at the top of the window while scrolling.
- Navigation links show an underline and a colour change on hover using border-bottom and transition.
- The Get Started button grows slightly on hover with transform: scale(1.08).

4.4.3 Output

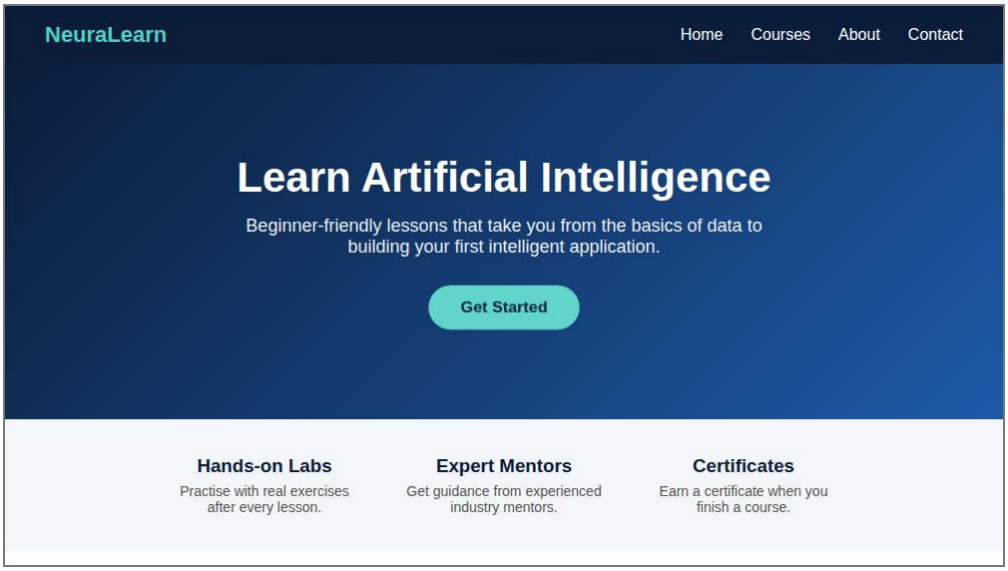


Fig. 4.5 – Landing page output (Get Started button in hover state)

4.5 Task 5: AI Chatbot Interface

Concepts used: Flexbox column layout, align-self, rounded chat bubbles, scrollable area, styled input bar.

4.5.1 HTML and CSS Code

File: task5_chatbot.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>AI Chatbot Interface</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; font-family: Arial, sans-serif; }
  body { background: #dfe6ee; display: flex; justify-content: center;
         align-items: center; min-height: 100vh; }
  .chat { width: 420px; height: 560px; background: #fff; border-radius: 16px;
          display: flex; flex-direction: column; overflow: hidden;
          box-shadow: 0 10px 30px rgba(0, 0, 0, 0.25); }
  .top { background: #2b6cb0; color: #fff; padding: 14px 18px; display: flex;
         align-items: center; gap: 12px; }
  .avatar { width: 38px; height: 38px; border-radius: 50%; background: #fff;
            color: #2b6cb0; display: flex; align-items: center;
            justify-content: center; font-weight: bold; }
  .top small { display: block; font-size: 12px; opacity: 0.85; }
  .messages { flex: 1; padding: 16px; background: #f5f8fc; overflow-y: auto;
              display: flex; flex-direction: column; gap: 12px; }
  .msg { max-width: 75%; padding: 10px 14px; border-radius: 16px;
         font-size: 14px; line-height: 1.45; }
  .bot { background: #fff; color: #222; align-self: flex-start;
         border-bottom-left-radius: 4px; box-shadow: 0 1px 3px rgba(0, 0, 0, 0.15); }
  .user { background: #2b6cb0; color: #fff; align-self: flex-end;
          border-bottom-right-radius: 4px; }
  .input-bar { display: flex; gap: 8px; padding: 12px; border-top: 1px solid #ddd; }
  .input-bar input { flex: 1; padding: 11px 14px; border: 1.5px solid #ccc;
                     border-radius: 22px; font-size: 14px; }
  .input-bar input:focus { outline: none; border-color: #2b6cb0; }
  .input-bar button { background: #2b6cb0; color: #fff; border: none;
                      border-radius: 22px; padding: 0 20px; cursor: pointer; }
  .input-bar button:hover { background: #1e4e8c; }
</style>
</head>
<body>
  <div class="chat">
    <div class="top">
      <div class="avatar">AI</div>
      <div>AI Assistant<small>&#9679; Online</small></div>
```

```
    </div>
    <div class="messages">
      <div class="msg bot">Hello! I am your AI assistant. How can I help you today?</div>
      <div class="msg user">What is the difference between HTML and CSS?</div>
      <div class="msg bot">HTML defines the structure and content of a web page, while
        CSS controls how that content looks &ndash; colours, fonts and layout.</div>
      <div class="msg user">Thanks, that was clear!</div>
      <div class="msg bot">You are welcome. Feel free to ask me anything else.</div>
    </div>
    <div class="input-bar">
      <input type="text" placeholder="Type your message...">
      <button>Send</button>
    </div>
  </div>
</body>
</html>
```

4.5.2 Explanation

- The .chat card is a Flexbox column; the .messages area uses flex: 1 so that it takes all the free space between the title bar and the input bar.
- align-self: flex-start places bot messages on the left and align-self: flex-end places user messages on the right.
- overflow-y: auto lets the message area scroll when the conversation grows longer.

4.5.3 Output

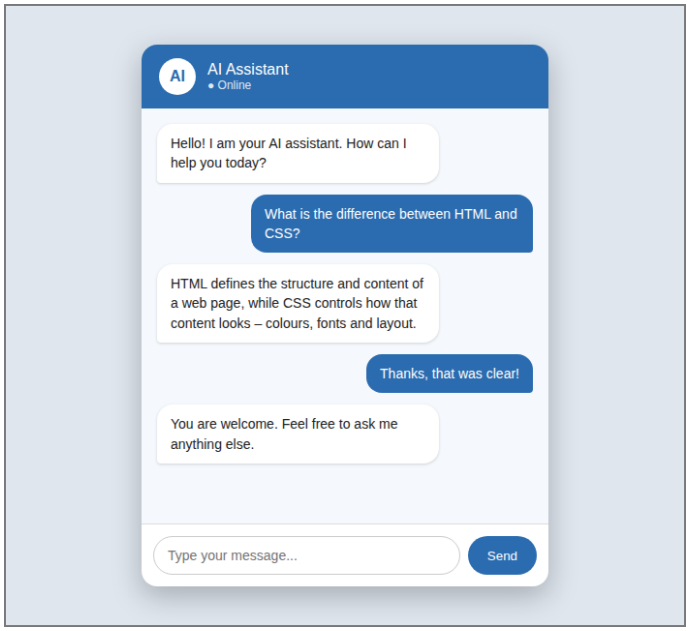


Fig. 4.6 – AI chatbot interface output

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Task 1 – Portfolio Page

The portfolio page displayed correctly in the browser. The header, three section cards and footer appeared in the intended order, and the skill tags wrapped onto new lines when the window was made narrower. Using semantic elements made the structure of the code easy to read.

Common Issues and Fixes

- Elements grew wider than expected when padding was added; setting box-sizing: border-box fixed this.
- Default browser margins created unwanted gaps around the header; the CSS reset removed them.

5.2 Task 2 – Registration Form

The form was centred on the page and every control was readable and clearly labelled. When a field was selected, its border changed colour and a soft glow appeared, which told the user where they were typing. The browser also checked the required and email fields before submission without any extra code.

Common Issues and Fixes

- Inputs overflowed the form width until width: 100% was combined with box-sizing: border-box.
- Clicking a label did not select its input until the for attribute matched the input id exactly.

5.3 Task 3 – Responsive Card Layout

On a wide screen the four cards appeared in one row. When the window was reduced below 800 px the cards moved into two columns, and below 480 px into a single column, as shown in Fig. 4.3 and Fig. 4.4. The hover effect lifted each card slightly and strengthened its shadow.

Screen width	Columns	Rule applied
Above 800 px	4	repeat(4, 1fr)
481 px to 800 px	2	@media (max-width: 800px)
480 px and below	1	@media (max-width: 480px)

Common Issues and Fixes

- The layout did not shrink on a phone until the viewport meta tag was added.

- The media query for the smaller screen had to be written after the larger one so that it overrides it correctly.

5.4 Task 4 – Landing Page

The navigation bar stayed at the top while scrolling, the links showed an underline and colour change on hover, and the hero section displayed a smooth blue gradient with a centred heading and button. The Get Started button enlarged and changed colour when the mouse pointer was placed over it.

Common Issues and Fixes

- The navigation links jumped by a few pixels on hover until a transparent border was reserved for them in the normal state.
- Transitions did not run when the property was missing from the transition list; listing the changed properties fixed this.

5.5 Task 5 – Chatbot Interface

The chatbot window displayed a title bar with an avatar and an online indicator, alternating bot and user messages, and an input bar with a Send button. Bot messages appeared on the left in white bubbles and user messages on the right in blue bubbles, which made the conversation easy to follow.

Common Issues and Fixes

- Messages stretched across the full width until max-width was set on each bubble.
- The input bar moved when the conversation grew, until the message area was given flex: 1 and overflow-y: auto.

Overall, all five pages rendered as intended in the browser and met the objectives of the internship tasks. Building them in sequence showed how the same few concepts, the box model, Flexbox, Grid and media queries, combine to create very different interfaces.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion of Individual Tasks

- **Task 1:** A clean portfolio page was built with semantic HTML5 and CSS cards, giving a solid understanding of page structure and the box model.
- **Task 2:** A user-friendly registration form was created with labelled inputs, built-in validation attributes and focus effects.
- **Task 3:** A responsive card layout was created with CSS Grid and media queries that adapts from four columns to one.
- **Task 4:** A landing page was created with a sticky navigation bar, a gradient hero section and smooth hover transitions.
- **Task 5:** An AI chatbot interface was designed with Flexbox, showing how the front end of an AI application can be built with only HTML and CSS.

6.2 Overall Conclusion

Across these five tasks, we progressed from a simple static page to a responsive, interactive-looking interface for an AI application. This progression demonstrated a consistent theme: well-structured HTML combined with well-organised CSS produces pages that are readable, maintainable and adaptable to any screen size. Together, these tasks gave us hands-on experience with semantic HTML5, form controls, the CSS box model, Flexbox, CSS Grid, media queries, gradients and transitions, which form the foundation for the user interface of any AI development project.

6.3 Future Work

- Add JavaScript to the chatbot interface so that messages can be typed, sent and displayed, and connect it to an AI model through an API.
- Add client-side validation and a back-end service to store the registration form data.
- Extend the landing page into a multi-page website with a responsive mobile navigation menu.
- Use CSS variables and a shared stylesheet so that all pages follow one consistent design.

INTERNSHIP CERTIFICATE

JARGON HANDLERS

Date: **08/09/2026**

TO WHOM SO EVER IT MAY CONCERN

This is to certify that Mr **M. Sushit** has undergone an Internship program in "**AI Development**" at Jargon Handlers Software Solutions Pvt Ltd.

The Internship Program was conducted from **August 1st, 2026** to **August 31st, 2026**. During this Internship Period, He has successfully completed his training as "**AI Developmen Intern**" .

We wish him every success in his life and career.

Best Regards,
For Jargon Handlers Software Solutions Pvt Ltd

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